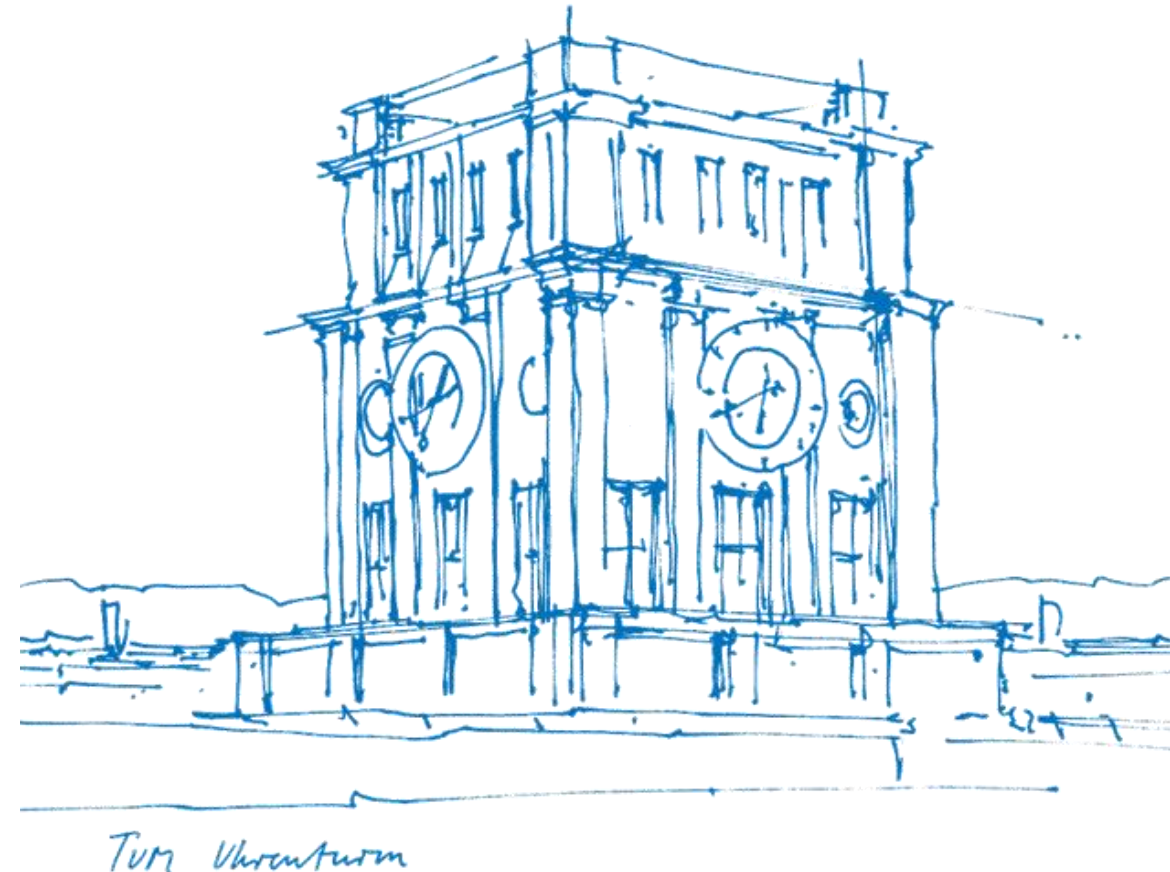


Tutorial 11

Maximum Likelihood Estimation (mle), Statistical Model

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About Tutorial

- **Tutorial:** In-Person, 2h by two time periods per week
 - 12:00-14:00 Tue, 03.11.018 (Group B33)
 - 10:00-12:00 Fri, 01.06.011 (Group E24)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Materials of this tutorial:** Please scan the QR-Code 

- **PART I.** Recap of the Lecture
- **PART II.** Exercises from the Exercise Sheet
- **PART III.** Additional Exercise (with solution)



I. Recap

• Maximum Likelihood Estimation (mle)

- ① - For dataset $x_1, x_2, \dots, x_n$, assume they come from a distribution with parameter θ .
 MLE finds the value $\hat{\theta}$ makes the observed data most likely, $L(\theta|x)$ achieves maximum value.
 $\hookrightarrow \hat{\theta}_{MLE} = \arg\max L(\theta|x)$.
 In that $L(\theta|x) = p(x_1, x_2, \dots, x_n|\theta)$ "Likelihood of θ given x ".
 e.g. Flipping coin (not average) 10 times, 7 heads, 3 tails.
 To estimate: "Prob. of head". $L(\theta) = \theta^7 (1-\theta)^3$
 Find $\hat{\theta}$ such that $L(\theta)$ gets the maximum!
 $\Rightarrow \hat{\theta} = 0.7$

- ② - Properties of estimators
- Bias (and unbiased estimator)
 - Mean squared error (mse)

• Statistical Model

- ③ - Definition

II. Exercise Sheet



Exercise 1 ([repeated from sheet 9:] mle for uniform distribution). Let $X_1, \dots, X_n$ be iid RVRVs from a uniform distribution $\text{Unif}(0, \theta)$ with unknown parameter $\theta > 0$. The pdf of X_i is $p(x_i | \theta) = \frac{1}{\theta} \chi_{[0, \theta]}(x_i)$.

- (i) Write down the likelihood function $L(\theta | x)$ for an observed sample $x = (x_1, \dots, x_n)$. Be careful with the indicator functions.
- (ii) Find the maximum likelihood estimator (mle) $\hat{\theta}_{\text{mle}}$ for θ .
[Hint: The likelihood function may not be differentiable with respect to θ , but the maximum value can be found by reasoning about monotonicity and where it is non-zero.]
- (iii) Calculate the expected value $\mathbb{E}[\hat{\theta}_{\text{mle}}]$. Is $\hat{\theta}_{\text{mle}}$ an unbiased estimator for θ ?
[Hint: First find the cdf of $\hat{\theta}_{\text{mle}}$, then its pdf, and then compute the expectation.]

II. Exercise Sheet



Exercise 2 (mle for exponential distribution). Let $X_1, \dots, X_n$ be iid RRVs from an exponential distribution for $n \in \mathbb{N}_{>1}$ with unknown rate parameter $\lambda > 0$, i.e., $X \sim (\text{Exp}(\lambda))_{\lambda \in \mathbb{R}_{>0}}$.

- (i) Write down the likelihood function $L(\lambda \mid x)$ and the log-likelihood function $\ell(\lambda \mid x)$ for an observed sample $x = (x_1, \dots, x_n)$ when given only the pdf of an $\text{Exp}(\lambda)$ distribution.
- (ii) Find the maximum likelihood estimator $\hat{\lambda}_{\text{mle}}$ for λ .
- (iii) Calculate the expected value $\mathbb{E}[\hat{\lambda}_{\text{mle}}]$. Is $\hat{\lambda}_{\text{mle}}$ an unbiased estimator for λ ?
[Hint: Since $X_i \sim \text{Exp}(\lambda) = \Gamma(1, \lambda)$, you can use that $S_n = \sum_{i=1}^n X_i \sim \Gamma(n, \lambda)$. Recall that $1/\bar{X}_n = n/S_n$ and use that for $Y \sim \Gamma(\alpha, \beta)$ we have that $\mathbb{E}[1/Y] = \beta/(\alpha - 1)$ if $\alpha > 1$.]

II. Exercise Sheet



Exercise 4. You're throwing $n \in \mathbb{N}$ darts at a target. Let $(X_i)_{i \in [n]}$ be iid random variables with unknown mean μ and unknown standard deviation σ , representing the distance of the i -th arrow from the center of the target. Furthermore, let $\hat{\mu} = \sum_{i=1}^n \lambda_i X_i$ be a convex combination of the individual X_i (i.e., $\lambda_i \geq 0$ for all $i \in [n]$ and $\sum_{i=1}^n \lambda_i = 1$).

- (i) Show that $\hat{\mu}$ is an unbiased estimator for μ .
- (ii) Prove that $\frac{1}{n} \sum_{i=1}^n \lambda_i^2 \geq \frac{1}{n^2}$.
[Hint: You can do this quickly using Jensen's inequality for an appropriate discrete random variable.]
- (iii) Determine the coefficients λ_i that minimize the mse of the estimator $\hat{\mu}$.

II. Exercise Sheet



Exercise 5. You have a box with $n \in \mathbb{N}_{>1}$ toasters among which a fixed number $d \in \{0, \dots, n\}$ is broken. You draw two toasters uniformly at random from the box.

- (i) What is the probability distribution P_d (depending on d) of the number of broken toasters among the two selected ones?

[Hint: Don't overcomplicate it, this is going to be a distribution over a set with just 3 elements. Feel free to specify the distribution via its pmf.]

- (ii) Assume d to be unknown and we want to estimate it from a (uniform) sample of two toasters. Write down an adequate statistical model and briefly explain it.
- (iii) Provide an unbiased estimator for d in this model and justify your choice.

II. Exercise Sheet

Exercise 3. Let X be a RVRV with pdf

$$p_X(x) = \begin{cases} 2c^2/x^3 & \text{if } x \geq c, \\ 0 & \text{otherwise,} \end{cases}$$

with $c > 0$.

- (i) Show that p_X is a valid pdf for all $c > 0$.
- (ii) Determine the cdf of X .
- (iii) Let $X_1, \dots, X_n$ be iid RVRVs from this distribution. Find the mle for c .
[Hint: For the likelihood function, consider the case where a sample value is less than c .]

III. Additional Exercise

Exercise 6. Let $X_1, \dots, X_n$ be independent samples of a random variable X . Determine the maximum likelihood estimator (MLE) for the respective parameter of the distribution under the given assumption. Investigate in both cases whether the estimator is unbiased.

- (a) X is Poisson-distributed with parameter $\lambda > 0$.
- (b) X is geometrically distributed with parameter $p \in (0, 1)$.

III. Additional Exercise

(a) $X \sim \text{Poi}(\lambda > 0)$. $P(X=n) = \frac{\lambda^n e^{-\lambda}}{n!}$, für unabhängige $x_1, x_2, \dots, x_n$ gilt:

$$\text{Likelihood-funk: } L(\vec{x}; \lambda) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

$$\ln L(\vec{x}; \lambda) = \sum_{i=1}^n \ln \left(\frac{\lambda^{x_i} e^{-\lambda}}{x_i!} \right) = \sum_{i=1}^n (x_i \ln \lambda - \lambda - \ln(x_i!)).$$

Ableiten nun: $(\ln L(\vec{x}; \lambda))' = \frac{\sum_{i=1}^n x_i}{n} - n = 0$, $\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n x_i$.

$E(\hat{\lambda}) = E\left(\frac{1}{n} \sum_{i=1}^n x_i\right) = \frac{1}{n} \sum_{i=1}^n \lambda = \lambda$, ist erwartungstreu.

(b) X geometrisch-verteilt. $P(X=n) = (1-p)^{n-1} p$.

$$\text{Likelihood-funk: } L(\vec{x}; p) = \prod_{i=1}^n (1-p)^{x_i} p.$$

$$\ln L(\vec{x}; p) = \sum_{i=1}^n \ln((1-p)^{x_i} p) = \left(\sum_{i=1}^n x_i\right) \ln(1-p) + n \ln p.$$

Ableiten nun: $(\ln L(\vec{x}; p))' = \sum_{i=1}^n \frac{x_i}{1-p} (-1) + \frac{n}{p} = 0$, $\hat{p} = \frac{n}{n + \sum_{i=1}^n x_i}$.

$E(\hat{p}) = E\left(\frac{n}{n + \sum_{i=1}^n x_i}\right) = p$, ist erwartungstreu.